# Humberto Sandmann

*Computer Scientist & Professor*

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São Paulo, Brazil | Open to relocation (on-site)

## PROFILE

Computer scientist and professor with a D.Sc. from the University of São Paulo and a research stay at the Max Planck Institute for Dynamics and Self-Organization. My research bridges bio-inspired neural computing, nonlinear dynamics, and pattern recognition, investigating how biological neuronal mechanisms can inform computational intelligence. More than a decade of university teaching alongside senior industry practice in software architecture, artificial intelligence, and high-performance computing.

## RESEARCH INTERESTS

- DeFi, smart contracts, and decentralized applications
- Small-world networks and complex systems
- Computation in large-scale systems
- Artificial intelligence and deep learning
- Nonlinear systems and network dynamics
- Synchronization of coupled oscillators

## EDUCATION

**D.Sc., Electrical Engineering** 2007–2012
*Polytechnic School, University of São Paulo*
*Thesis* Spike patterns and computation in dynamical neural networks
*Advisor* Prof. Dr. Emílio Del Moral Hernandez
*DOI* 10.11606/T.3.2012.tde-05092012-165022

**D.Sc. Research Internship** 2009–2010
*Max Planck Institute for Dynamics and Self-Organization · Göttingen, Germany*
*Advisor* Prof. Dr. Marc Timme
*Note* Networks Dynamics Group — CAPES grant BEX 4076-09-5

**M.Sc., Electrical Engineering** 2003–2006
*Polytechnic School, University of São Paulo*
*Thesis* Prediction of time series using architecture based on neuro-fuzzy systems
*Advisor* Prof. Dr. Marco Túlio Carvalho de Andrade
*DOI* 10.11606/D.3.2006.tde-01042009-095125

**B.Sc., Computer Science** 1999–2002
*Faculdade de Engenharia Industrial (FEI) · São Bernardo do Campo, Brazil*

**Technician, Industrial Computation** 1996–1998
*Escola Técnica Jorge Street · São Caetano do Sul, Brazil*

## PUBLICATIONS

H. Sandmann, L. A. Silva, E. Del-Moral-Hernandez. **A self-organizing architecture of recursive elements for continuous learning (poster abstract).** *Dynamics Days Europe, Göttingen, Germany*, 2009, pp. 246–248.

E. Del-Moral-Hernandez, H. Sandmann, G. Araujo. **Context dependent pattern recognition — A framework for hybrid architectures bridging chaotic neural networks based on Recursive Processing Elements and symbolic information.** *Proc. International Joint Conference on Neural Networks (IJCNN), Atlanta, GA, USA*, 2009, pp. 663–670. doi:10.1109/IJCNN.2009.5179061

L. A. da Silva, H. Sandmann, E. Del-Moral-Hernandez. **A self-organizing architecture of recursive elements for continuous learning.** *Proc. IJCNN (IEEE World Congress on Computational Intelligence), Hong Kong, China*, 2008. doi:10.1109/IJCNN.2008.4634190

H. Sandmann, M. T. C. de Andrade. **Análise da capacidade de uma rede neural de memória associativa com arquitetura de elementos processados recursivamente.** *Proc. Brazilian Symposium on Neural Networks (SBRN), São Luís, MA, Brazil*, 2004.

E. Del-Moral-Hernandez, H. Sandmann, L. A. da Silva. **Pattern recovery in networks of recursive processing elements with continuous learning.** *Proc. IJCNN, Budapest, Hungary*, 2004. doi:10.1109/IJCNN.2004.1379877

## ACADEMIC APPOINTMENTS & TEACHING

**Insper** — *Associate Professor, Assistant Professor* 2018–present
*Platform, Microservices and APIs · Machine Learning · Neural Networks and Deep Learning · Cloud Computing · Design of Software · Coding Interviews · Foundations for Physical Communication*

**ESPM** — *Associate Professor* 2014–2025
*Machine Learning · Applied Mathematics · Object-Oriented Programming · Software Quality · Statistics · Architecture and Operating Systems · Artificial Intelligence and Society*

**FIAP** — *Associate Professor* 2012–2015
*Distributed Systems · MBA: Big Data & NoSQL Modelling · Data Structures and Algorithms · Databases (Oracle PL/SQL)*

**Toledo Prudente Centro Universitário** — *Visiting Professor* 2015–2015
*MBA: Big Data and Machine Learning*

**Universidade Cruzeiro do Sul** — *Associate Professor* 2012–2013
*Computer Architecture · Software Engineering · Mobile Programming (Android) · Databases*

**Universidade Nove de Julho** — *Temporary Professor* 2011–2011
*Computer Architecture · Introduction to Programming · Web Programming*

**Polytechnic School, University of São Paulo** — *Teaching Assistant* 2008–2011
*Experimental Pattern Recognition, Modeling and Neurocomputing · Modeling for Signal Processing: Neural Networks and Learning · Foundations of Computing Engineering Laboratory*

## PROFESSIONAL EXPERIENCE

| | | |
|---|---|---|
| **H-Sandmann** | Founder | 2006–2025 |
| **SelsanTech** | Co-Founder | 2012–2023 |
| **RockSpoon** | Senior Software Architect & Machine Learning Engineer | 2018–2019 |
| **Banco Votorantim** | Senior Consultant, Computational Optimization | 2015–2016 |

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|---|---|---|
| **TecBan** | Senior Consultant, Computational Optimization | 2011–2011 |
| **ABN Amro Bank** | Senior Software Engineer / Architect | 2006–2007 |
| **Politec** | Senior System Analyst | 2005–2006 |
| **EverSystems** | Senior Software Engineer / Architect | 2004–2005 |
| **Central Online** | Senior System Analyst | 2002–2003 |
| **Datamace** | Senior System Analyst | 1999–2002 |
| **Heraeus Electro-Nite** | Computer Programmer | 1998–1999 |

## GRANTS & FUNDING

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|---|---|---|
| **CNPq** | Doctoral fellowship — grant 140632/2007-7 | 2007 – 2011 |
| **CAPES** | Doctoral internship abroad (Max Planck Institute) — BEX 4076-09-5 | 2010 |
| **CAPES (PAE)** | Teaching-assistantship fellowship — Polytechnic School, USP | 2008, 2011 |
| **FAPESP** | Teaching-assistantship fellowship — Polytechnic School, USP | 2008 |

## CONFERENCE SERVICE

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|---|---|---|
| **IJCNN 2020** | Reviewer | Glasgow, Scotland |
| **IJCNN 2019** | Program Committee & Reviewer | Budapest, Hungary |
| **IJCNN 2018** | Reviewer | Rio de Janeiro, Brazil |
| **IJCNN 2017** | Program Committee & Reviewer | Anchorage, AK, USA |
| **IJCNN 2015** | Reviewer | Killarney, Ireland |

## PROFESSIONAL MEMBERSHIPS

**IEEE — Institute of Electrical and Electronics Engineers** — Member #60216475

- Computational Intelligence Society (CIS) — member since 2004; Brazilian committee (2009; 2010–2012)
- Signal Processing Society (SPS) — 2005–2009
- Engineering in Medicine and Biology Society (EMBS) — 2005–2009

**INNS — International Neural Network Society** — Student member #3951

## LANGUAGES

**Portuguese** — Native **English** — Advanced **German** — B1